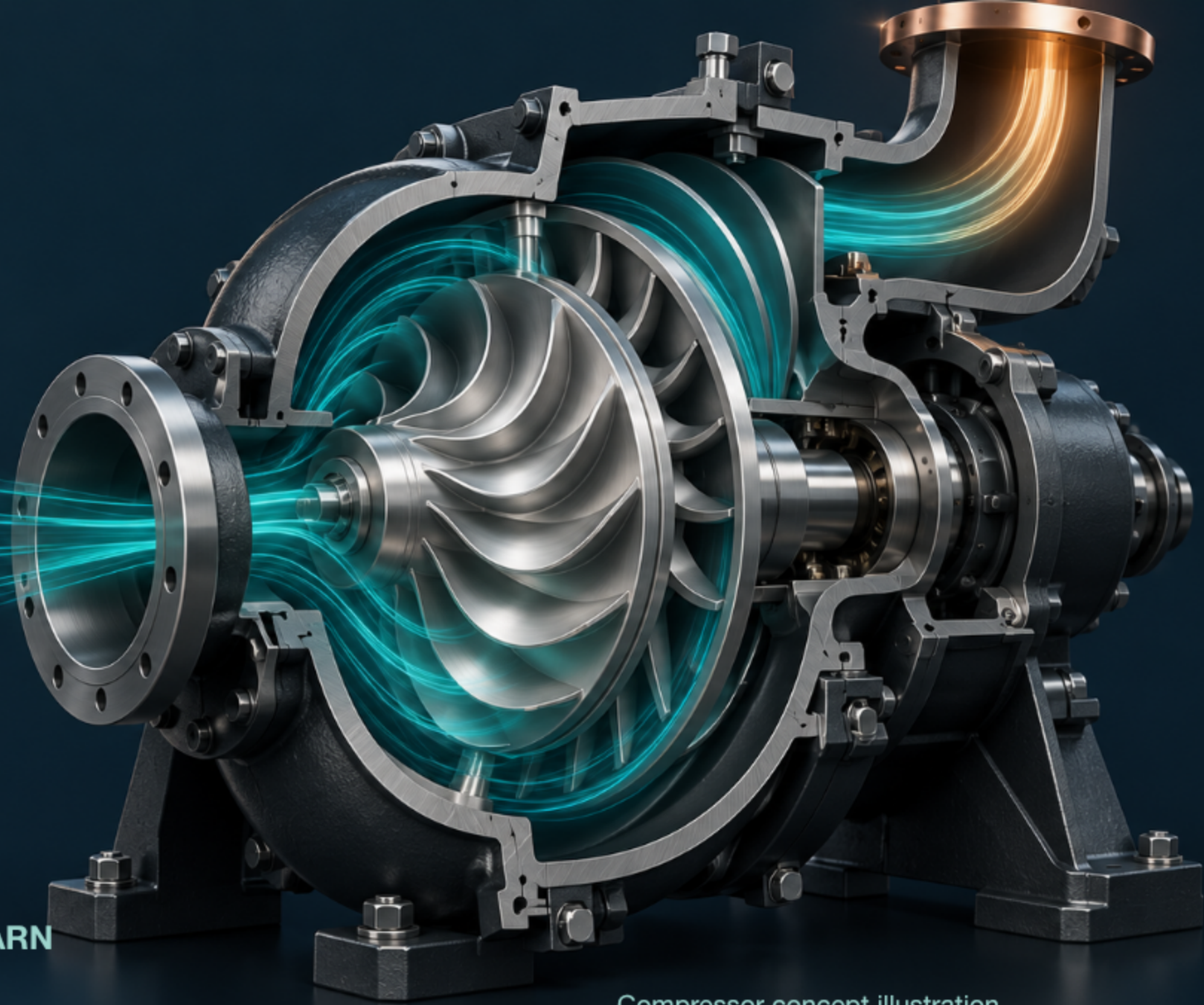


ACE

Learn what to change next.

Causal models for
smarter experiments

STRUCTURE → INTERVENE → LEARN



Compressor concept illustration

The causal model inside ACE

$$X_i = f_i(Pa_i, U_i)$$

Parents Pa_i supply inputs. U_i represents outside influences.



Worked example: known graph, unknown slopes, zero disturbances and dimensionless values.

Choose where to intervene

Ordinary system



Experiment: do(M = 1)



Incoming edge removed

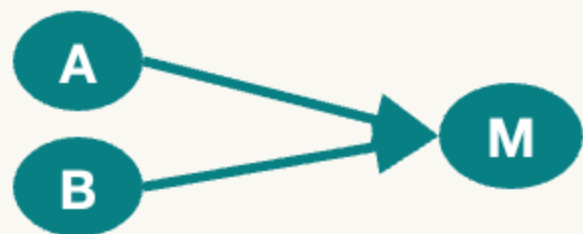
Y still responds to M

One paid response can train several natural mechanisms. It is still one paid response.

Local mechanism tables vs one joint table

Same system: 10 inputs, 5 settings each, 9 mechanisms with two parents

SCM LOCAL TABLES



A	B	M = f(A, B)
0	0	m_1
0	1	m_2
...	...	...
4	4	m_{25}

25 rows per mechanism $\times$ 9 = 225

Discrete illustration. Local maps compose through the graph. ACE fits functions instead of literal tables.

SCM-FREE JOINT TABLE

All input settings together determine Y

X_1	X_2	...	X_{10}	Y
0	0	...	0	y_1
0	0	...	1	y_2
...	...	...	...	...
4	4	...	4	y_n

$5^{10} = 9,765,625$ joint rows

Exhaustive grids multiply the work

10 inputs · 5 settings each · a deliberately naive joint-table baseline

SCM-FREE JOINT TABLE

9.77 million

configurations for exhaustive coverage

29.26 million random draws

for 95% expected joint-grid coverage¹

SCM LOCAL TABLES

225

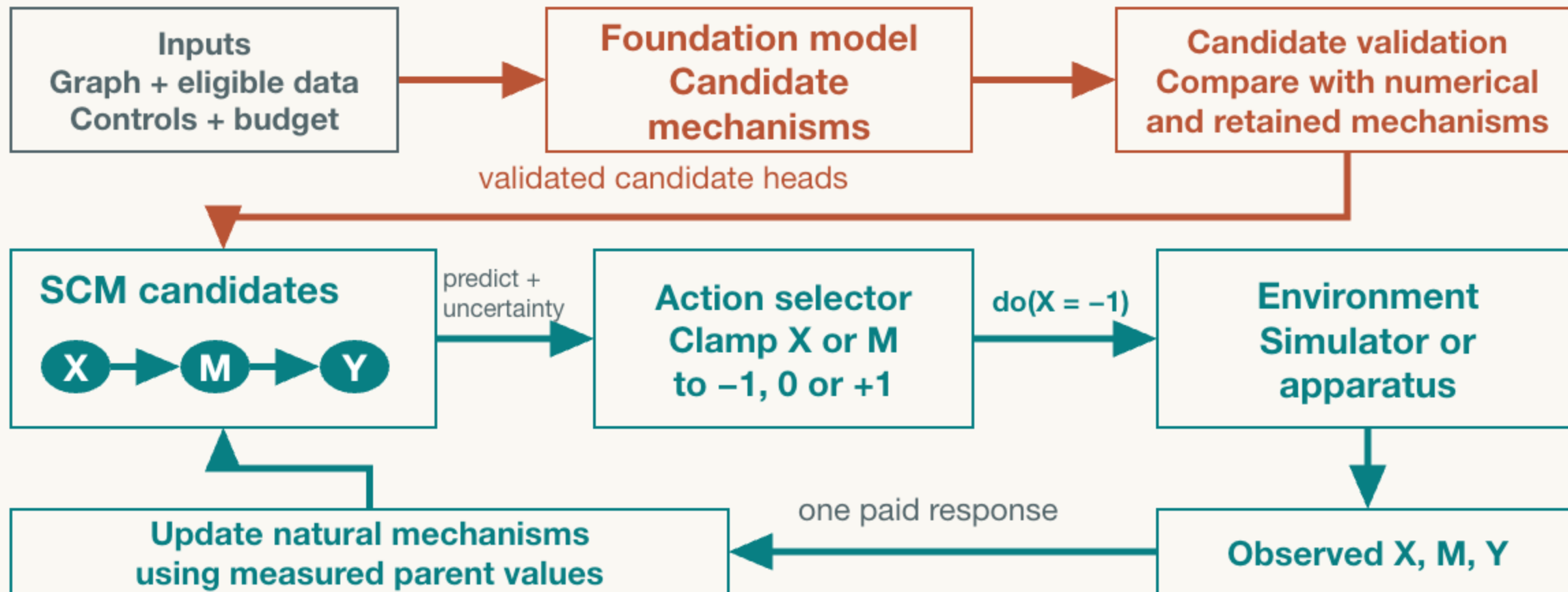
local entries

9 mechanisms
25 settings each

Analytic coverage / representation example, not a measured ACE speedup. ¹ Uniform draws with replacement.

ACE: models, actions and environmental feedback

Foundation-model branch: proposed integration



Exclude the clamped mechanism. Stop when the budget ends. Measured curves use the neural SCM loop.

Exact inputs to the three-node demonstration

Mechanism	Member slopes	Mean prediction
M from X	1, 2, 3	$\hat{M} = 2X$
Y from M	2, 3, 4	$\hat{Y} = 3M$

Actions: clamp X or M to -1, 0 or +1

References: -1, 0, +1 Noise proxy: 0.05
Budget: one response Ties: first action in the list

Simplified linear ensemble · six candidate actions · one response budget

The score selects $\text{do}(X = -1)$

Intervention	M contribution	Y contribution	Total
$\text{do}(X = -1)$	0.413437	0.436264	0.849701
$\text{do}(X = 0)$	0.000000	0.000000	0.000000
$\text{do}(X = 1)$	0.413437	0.436264	0.849701
$\text{do}(M = -1)$	0.000000	0.413437	0.413437
$\text{do}(M = 0)$	0.000000	0.000000	0.000000
$\text{do}(M = 1)$	0.000000	0.413437	0.413437

Selected prediction: $\hat{M} = -2$, $\hat{Y} = -6$

Score contribution: $\text{mean Cov}^2 / (\text{Variance} + \text{noise proxy}) \cdot \text{tied actions follow the declared order}$

The response updates eligible mechanisms

Observed row: $X = -1$, $M = -2.5$, $Y = -7.5$

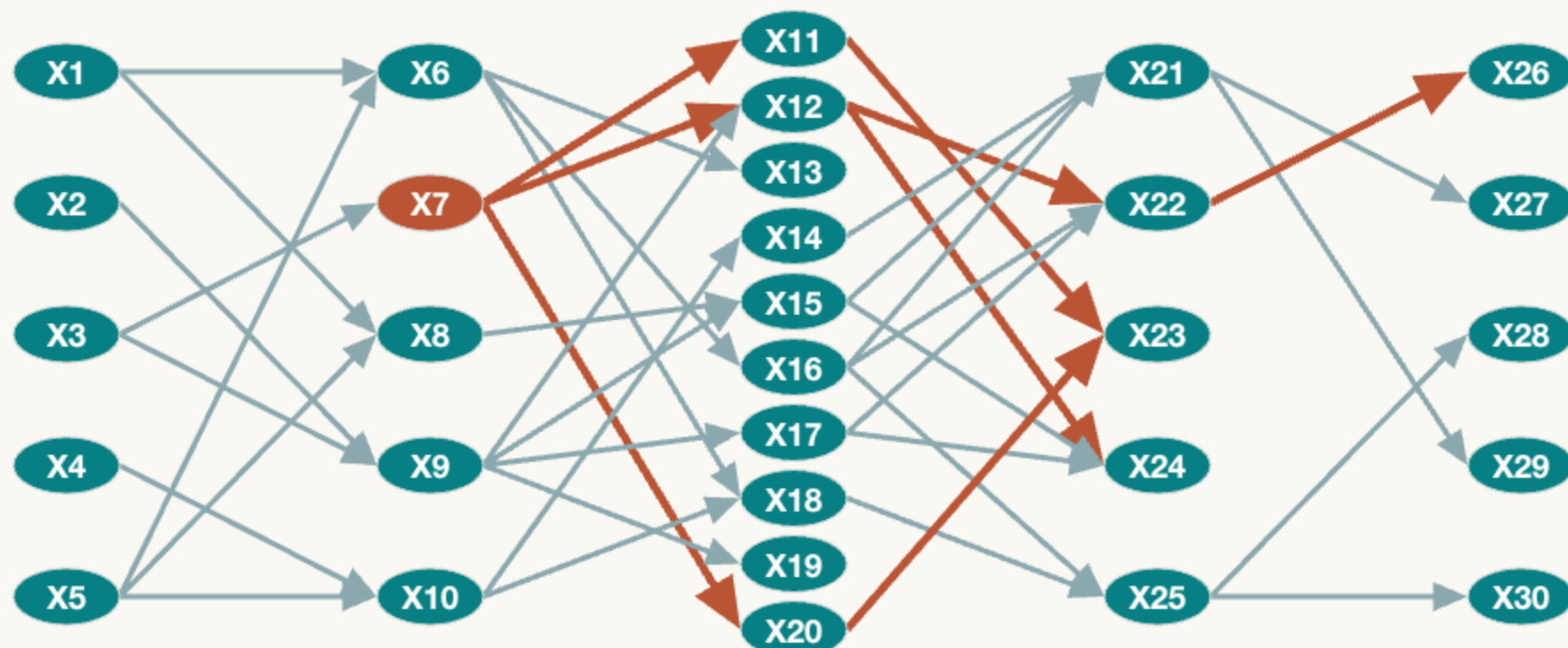
Natural mechanism	Training input	Training label
M	Measured $X = -1$	$M = -2.5$
Y	Measured $M = -2.5$	$Y = -7.5$

After one illustrative gradient update

$M = -2.25$ **$\hat{Y} = -6.75$** **Budget left: 0**

Illustrative linear update · both natural mechanisms learn from the same paid response

A real scenario: 30 linked mechanisms



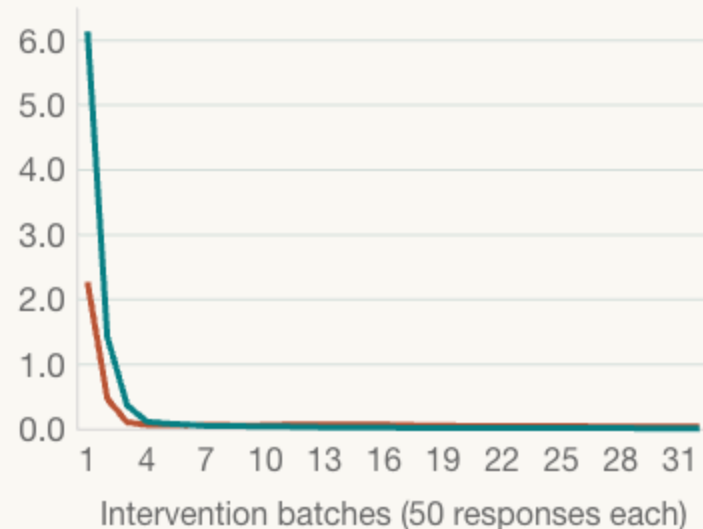
First selected action: $\text{do}(X7 = 3.8571)$ $\rightarrow$ 50 responses

Arrows point from cause to effect. Coral follows the X7 intervention downstream · archived seed 5000.

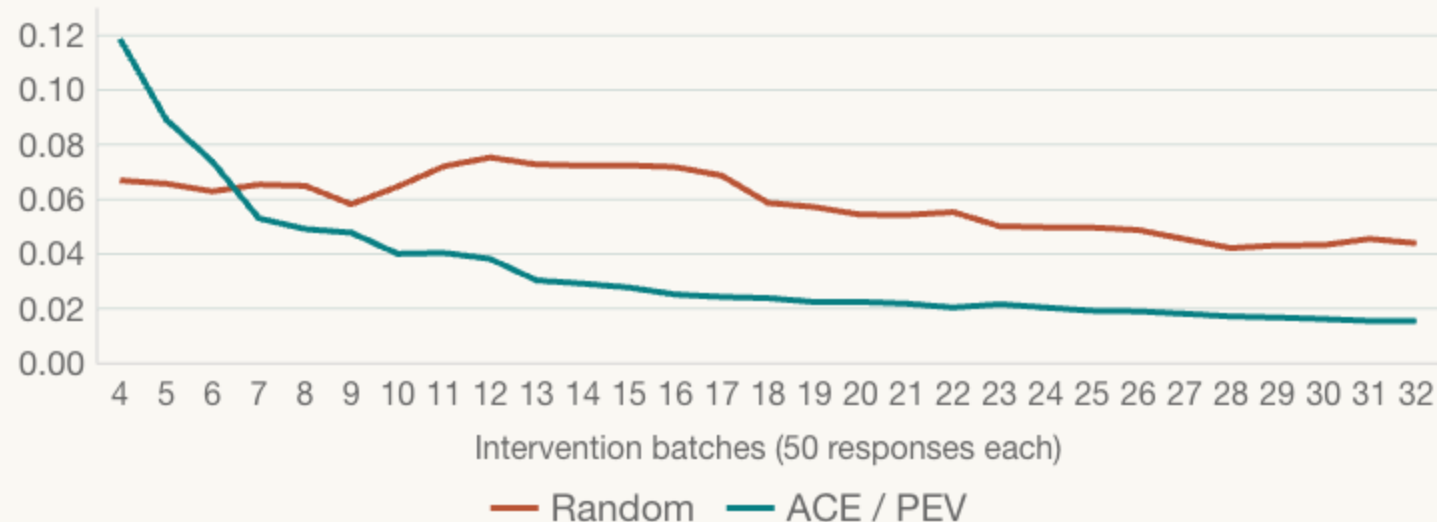
65% lower final prediction error

19 of 20 systems won · equal budgets · lower error is better

Full recorded curve



Detail: batches 4–32



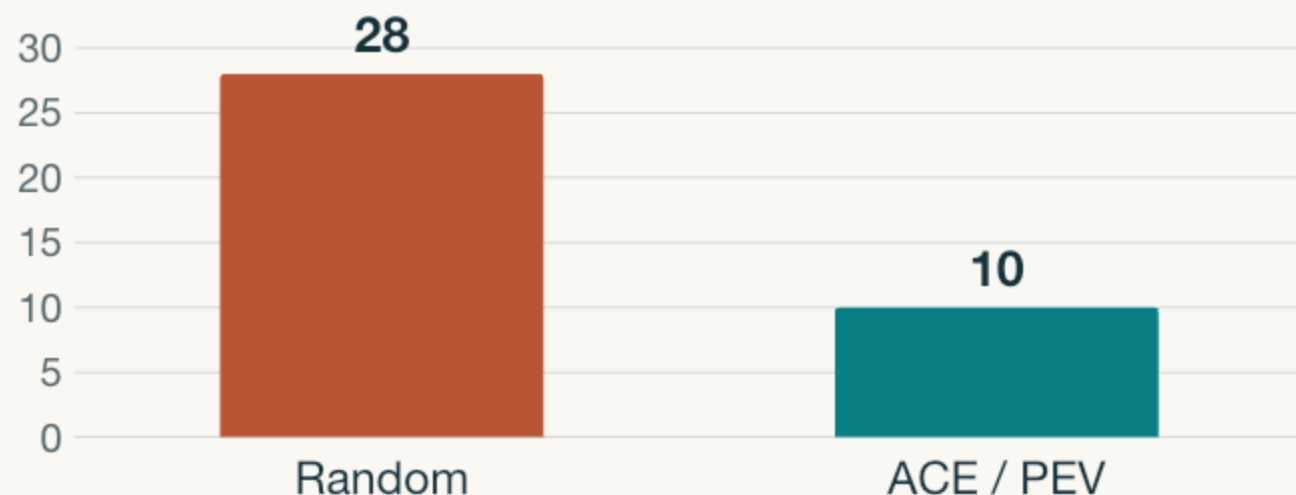
Final MSE: 0.01543 ACE vs 0.04390 random | 19 / 20 wins

20 synthetic systems · 2,000 responses per arm · random action selection with the same SCM learner

64% fewer intervention batches

MEASURED INTERVENTION BATCHES

First mean-curve crossing of $MSE \leq 0.04390$



64.3% fewer

**1,400 to 500
intervention responses**

50 responses / batch. Including observations: 1,760 to 620.

Retrospective curve comparison

SCM-FREE JOINT TABLE

10 inputs $\times$ 5 settings

9.77 million

entries for exhaustive coverage

Input settings	Y
0, 0, ..., 0	y_1
...	...
4, 4, ..., 4	y_n

**29.26 million random
draws**

95% expected grid coverage

Analytic illustration, separate units